Bravo 5: Cathode Protection

Carlos Suarez Zapico

March 2026

1 Neccesary Conditions for the arm

- Slip rings in continuous rotation joints to avoid cable twisting.
- Joint current control watchdog to prevent arm from going haywire if communication is lost.

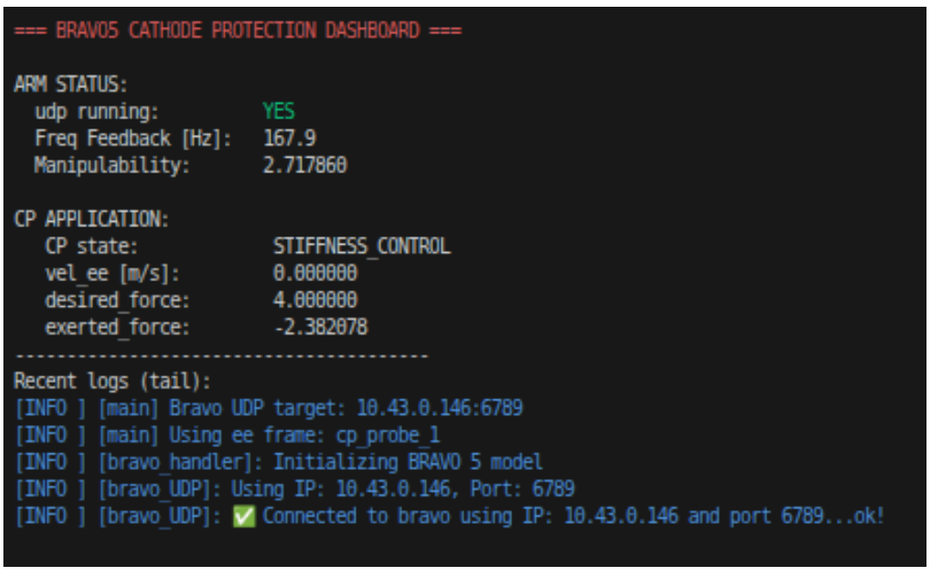

Figure 1: Terminal graphical interface for the arm and CP application.

$$\mathrm{F}_{\text{task}} = \mathbf{K}_x(\mathbf{x}_{\text{des}} - \mathbf{x}) + \mathbf{D}_x(\dot{\mathbf{x}}_{\text{des}} - \dot{\mathbf{x}})$$

$$\boldsymbol{\tau}_{\text{task}} = \mathbf{J}^\top \mathbf{F}_{\text{task}}$$

(1)

$$\mathbf{K}_x = \begin{bmatrix} k_x & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & k_z \end{bmatrix} \tag{2}$$

$$\ddot{\mathbf{x}}_{\text{des}} = \mathbf{M}^{-1}\left(\mathbf{F}_{\text{ext}} - \mathbf{D}_x\dot{\mathbf{x}} - \mathbf{K}_x(\mathbf{x} - \mathbf{x}_{\text{eq}})\right) \tag{3}$$

2 Stiction

2.1 Approach 1: Friction feedforward (with Stribeck)

Identify static friction torque τ_s per joint. Apply it in the command before feedback kicks in. For example:

$$\tau_{\text{cmd}} = \tau_{\text{ctrl}} + F_s \cdot \text{sgn}_\epsilon(\dot{q}) \tag{4}$$

where sgn_ϵ is a smooth sign function to avoid chatter. This lets you lower feedback gains but still overcome the threshold.

2.2 Approach 2

2.2.1 Separate loops + keep the outer one soft

- Inner loop: fast current (torque) control. Aim for high bandwidth and good disturbance rejection here.
- Outer loop: low-gain impedance/admittance layer that commands torque, not position. Use a virtual spring-damper: $\tau_c = K_p(q_{ref} - q) + K_d(\dot{q_{ref}} - \dot{q})$. Keep K_p,K_d modest for compliance, then add smart feedforward so you don't need big gains.

2.2.2 Put friction in the feedforward path (not the feedback)

Model + compensate for the stuff that's forcing you to raise gains. Simple friction model (good start):

$$\tau_f = F_c \mathrm{sgn}(\dot{q}) + B\dot{q} + (F_s - F_c)e^{-(\dot{q}/v_s)^2}\mathrm{sgn}(\dot{q}) \tag{5}$$

Then command:

$$\tau_{cmd} = \tau_c + \tau_f + \tau_g \tag{6}$$

where τ_g is gravity compensation (from your model or ID'd offline).

If you don't want to fit Stribeck yet, start with Coulomb + viscous (two parameters per joint). Identify by sweeping at slow velocities in both directions and solving for F_c and B. Implementation tips: Estimate $\dot{q}$ with a well-tuned low-pass differentiator/observer (e.g., 2nd-order Butterworth with cutoff comfortably above motion bandwidth, below current loop bandwidth). Blend friction near zero speed to avoid sign chatter:

$$\mathrm{sgn}_\epsilon(\dot{q}) = \frac{\dot{q}}{\sqrt{\dot{q}^2 + \epsilon^2}} \tag{7}$$

2.2.3 Use a tiny "dither" to break stiction (optional but effective)

Inject a small sinusoidal current at a few Hz (or just above your mechanical resonance) when $|\dot{q} > 0|$ and $qq > 0$. Amplitude: just enough to exceed stiction (a few percentage of rated torque). Gate it off once motion starts. This reduces stick-slip without raising gains.

- Add a small high-frequency torque (a few percentage of rated torque) around zero velocity.
- Frequency: above your motion bandwidth but below mechanical resonances (often 20–80 Hz for small arms).
- Effect: Keeps micro-motion at contact points so surfaces never "fully stick."

pseudo Copy Edit

```
loop at fast rate:
  q, dq = read_encoders_and_velocity()
  tau_g = gravity_comp(q)
  tau_ff = friction_ff(dq)          // Coulomb+viscous(+Stribeck), smoothed near 0
  tau_imp = Kp*(q_ref - q) + Kd*(dq_ref - dq)    // low gains for compliance

  if near_zero_speed_and_error_significant:
      tau_imp += dither(t)          // small sinusoid, gated

  tau_cmd = saturate(tau_imp + tau_ff + tau_g, tau_limits)
  i_cmd = tau_cmd / Kt              // current command via motor torque constant
  send_to_current_loop(i_cmd)
```

Figure 2: PseudoCode

3 Stribeck Effect

The Stribeck effect is a phenomenon in friction where the friction force decreases as speed increases from zero. It is basically the transition from "sticky" to "slippery."

3.1 What's going on physically?

At zero velocity, static friction (stiction) dominates — it takes a certain torque/force just to break loose. As soon as motion starts, the contact surfaces shift from "micro-welded" to "sliding with a thin lubricant film" or micro-rolling, so the required force drops — this is the Stribeck region. After that drop, viscous friction (proportional to speed) takes over, so total friction rises again at higher speeds.

3.2 Mathematical Model

$$F_f(\dot{q}) = \left[F_c + (F_s - F_c)e^{-(\dot{q}/v_s)^2}\right]\operatorname{sgn}(\dot{q}) + B\dot{q} \tag{8}$$

where:

- F_s static friction
- F_c Coulomb (constant sliding) friction
- v_s Stribeck velocity (where drop happens)
- B viscous coefficient

3.3 Why it matters for robot control

- If you ignore it, your low-speed torque commands will overshoot → jerky starts/stops.
- When trying to control low contact forces or track slow motions (especially in current control), Stribeck friction is the main reason you need higher gains to break stiction — and higher gains kill compliance.
- Modeling + feedforward compensating for it lets you run much lower position/impedance gains while still moving smoothly.

Stiction (static friction) is particularly nasty because:

- In **free motion**: you need a sudden torque jump to break loose → jerk, overshoot, buzzing.
- In **force control**: near zero velocity, the joint "sticks" until the commanded force exceeds static friction, then it "slips" and overshoots the target force — classic stick–slip instability.
- **Nonlinearity**: Unlike viscous friction (smooth, proportional to velocity), static friction is a discontinuous threshold. Your controller can't just scale output linearly — it's like pushing a stubborn door: nothing happens until suddenly everything moves.
- **Deadzone Effect**: If your desired torque is below (the static friction torque), nothing moves. This destroys your ability to apply gentle forces or track small motions.
- **Force ripple**: In contact, the "break–slip–stick" cycle makes your force measurement oscillate instead of staying constant.

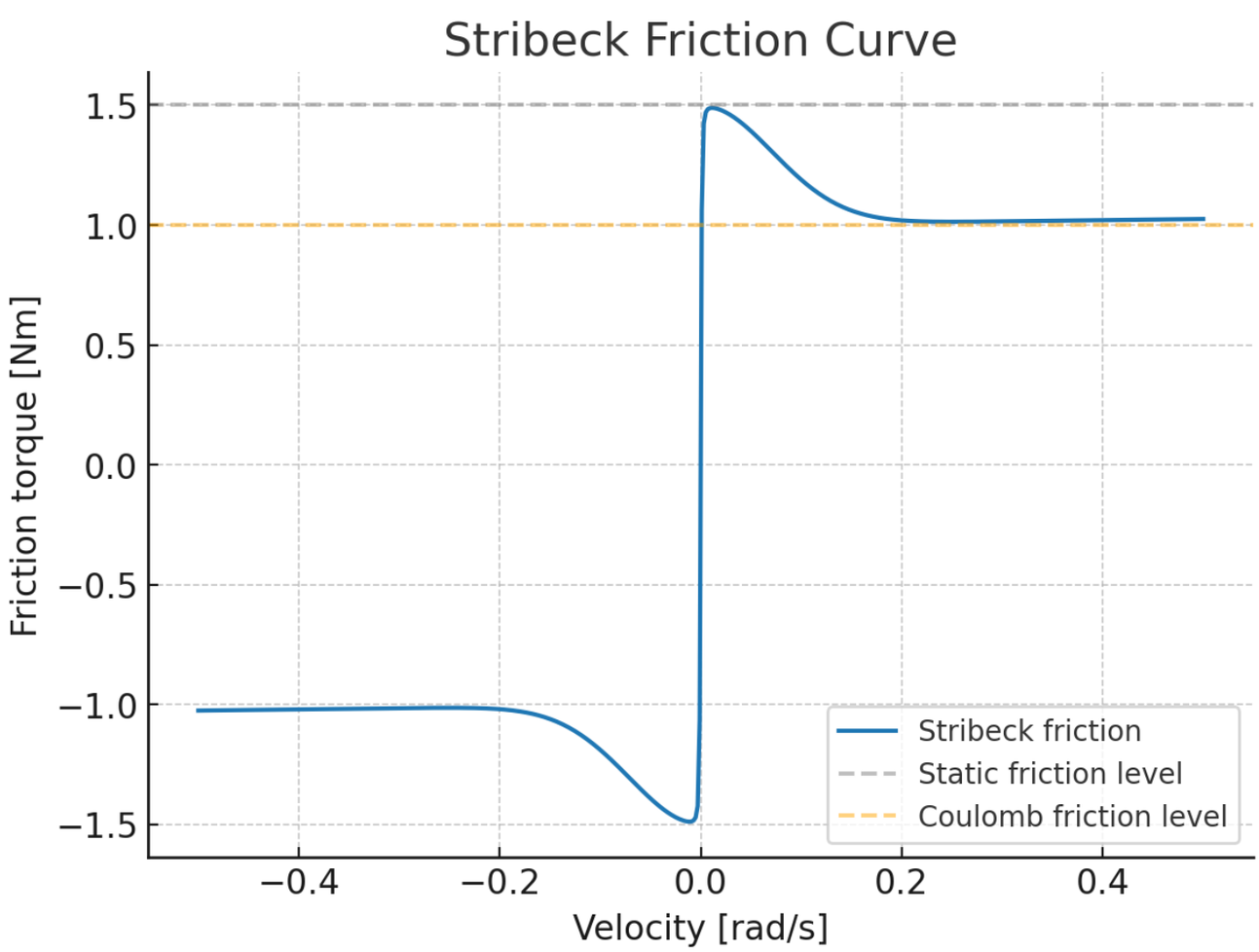

Figure 3: Stribeck

Here's the Stribeck curve — you can see the peak at zero speed (static friction), then the drop toward the sliding level as speed increases, and finally the viscous slope at higher velocities. This "sticky → slippery" transition is exactly what causes jerky starts and stick–slip.

4 Stiction